

E-R and EER Models

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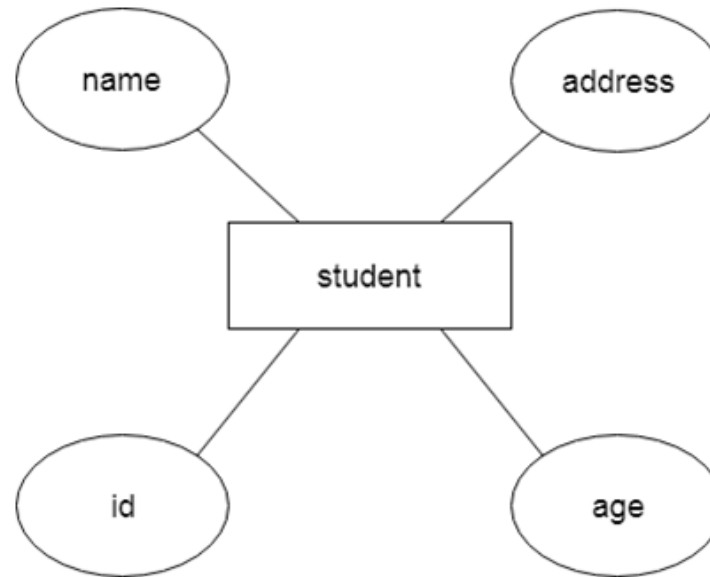
Reference Book: INTRODUCTION TO DATABASE MANAGEMENT SYSTEM
by Dr. Satinder Bal Gupta & Aditya Mittal

INDEX

- i. Introduction
- ii. Basic Concepts
- iii. Types of Attribute
- iv. Relationship Sets
- v. Mapping Constrains
- vi. Keys
- vii. Entity – Relationship Diagram
- viii. Types of Entity Sets
- ix. Enhanced Entity-Relationship (EER) Model
- x. Reduction of an E-R and EER Diagram into tables

Introduction

- Entity-Relationship model, is a high-level data model used in Database Management Systems to represent the logical structure of database.
- It is conceptual design that simplifies the of data by defining entities, attributes, and relationships between them.



Basic Concepts

- **Enterprise:** Enterprise refers to any kind of organization.
Ex. School, Colleges any company etc.
- **Entity:** Entity refers to an “object” or “thing” in real world. Object may be any person, place, event etc.
Ex. Students of colleges and schools, loans in banks, employees in any company etc.
- **Attributes:** These are the characteristics of any entity.
Ex. A student can be described by his name, age, address, height, class etc.
- **Value:** Value is the information or data which is stored in attributes of any entity.
- **Entity Sets:** All the entities having same attributes make an entity set.
- **Domain:** Domain or value set is the set of all values or information about any attribute.

Ex. Consider the student table shown in Figure 2.1. It describes the basic concepts.

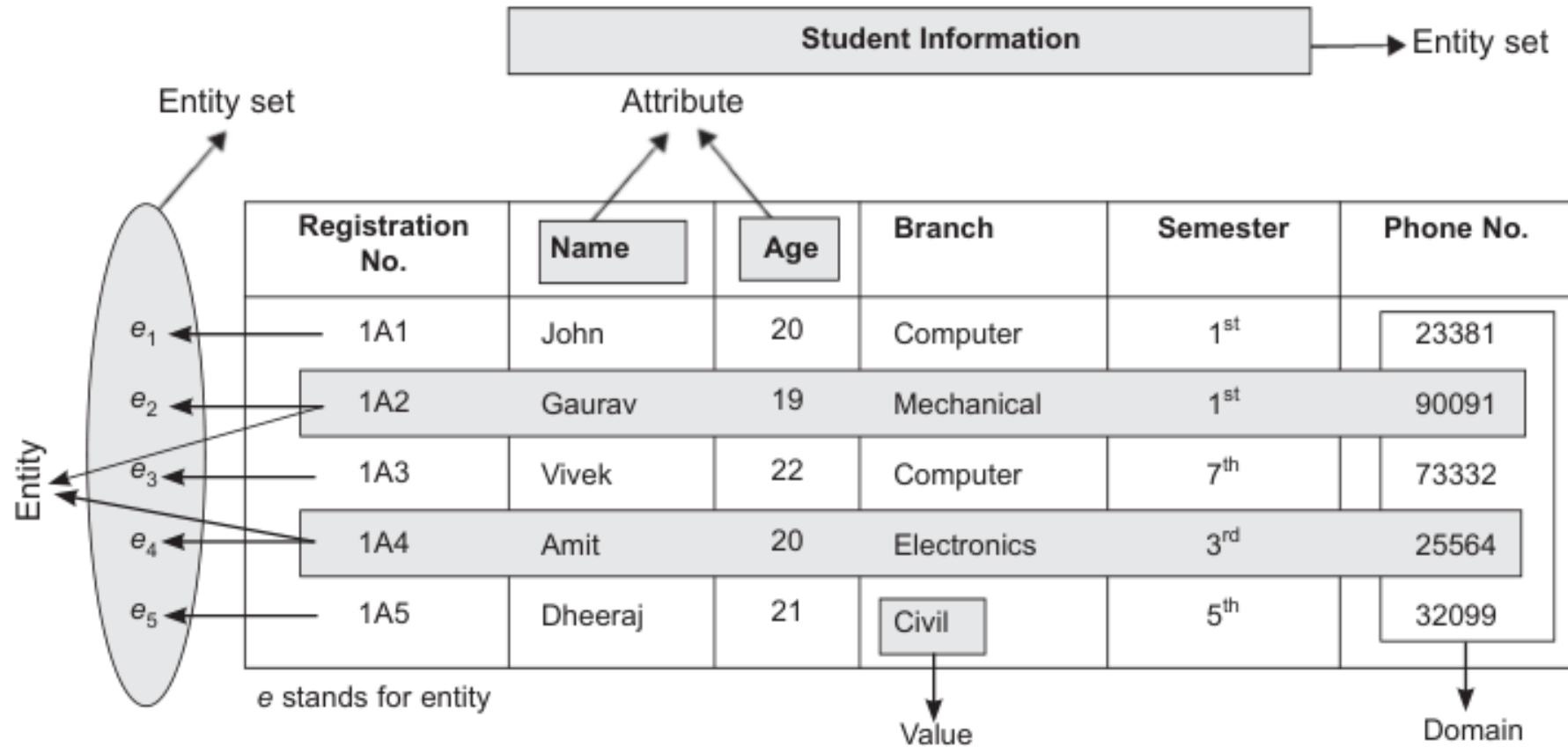


FIGURE 2.1. Table showing basic concepts of E-R model.

Types of Attributes

1. Simple and Composite Attribute

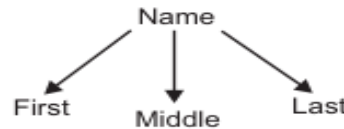
i) **Simple:** Attributes are those which cannot be divided into subparts.

Ex. Age of a student.



ii) **Composite:** Attributes are those which can be divided into subparts.

Ex. Name of a student can be divided into First Name, Middle Name, and Last Name.



2. Single Valued and Multi-valued Attribute

i) **Single Valued:** An attribute having only single value for a particular entity is known as single value attribute. *Ex. Age of a student.*

ii) **Multi-valued:** An attribute having more than one possible value for a particular entity is known as multi-valued attribute.

Ex. Phone number of a student. A student may have more than one phone.

3. Derived Attributes and Stored Attributes

i) **Derived Attributes** : An attribute that can be derived from other known attributes is known as derived attribute.

Ex. Age of employees can be derived if you know date of birth and system date.

$$\text{Age} = \text{System date} - \text{Date of birth}$$

ii) **Stored Attributes** : An attribute which cannot be derived by other known attributes is known as stored attribute.

Ex. Date of birth of any employee.

***NULL Value** : Null stands for nothing. An attribute have a null value if either the value of that attribute is not known or the value is not applicable.

Caution: NULL is not equal to Zero (0). But you can say that NULL is blank as shown in Figure 2.2.

Ex.

Name	Subject	Marks
abc	Maths	92
def	Science	—
ghi	Maths	0

→ This is null

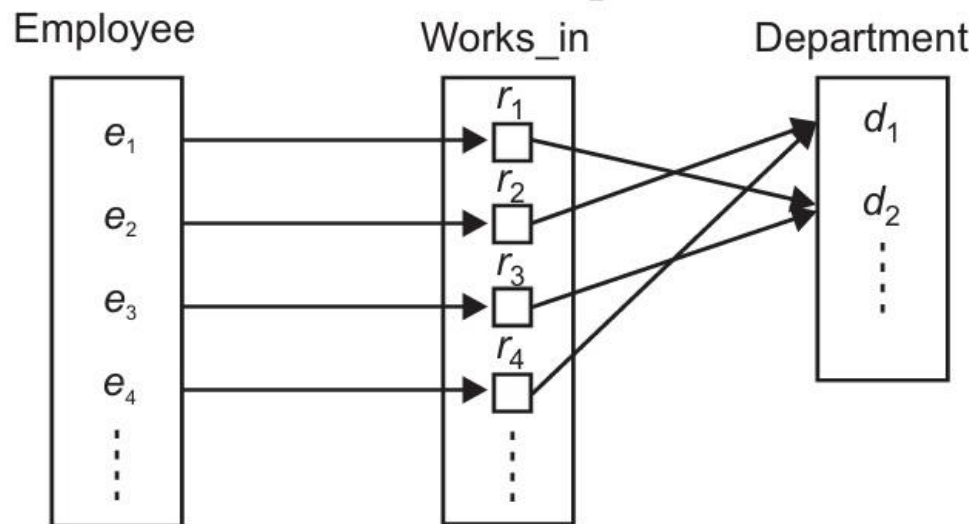
FIGURE 2.2. Showing null value.

Relationship Sets

a) **Relationship:** A relationship is the association among several entities. It connects different entities through a meaningful relation.

b) **Relationship Set:** A relationship set is a set of relationships of the same type.

- Ex. Employees work in different departments. Then relationship exists between employees and departments because each employee must belong to some department.
- Here, Employee and Department are two entity sets. r stands for relationship between Employee and Department. Works_in is the relationship set as shown in Figure



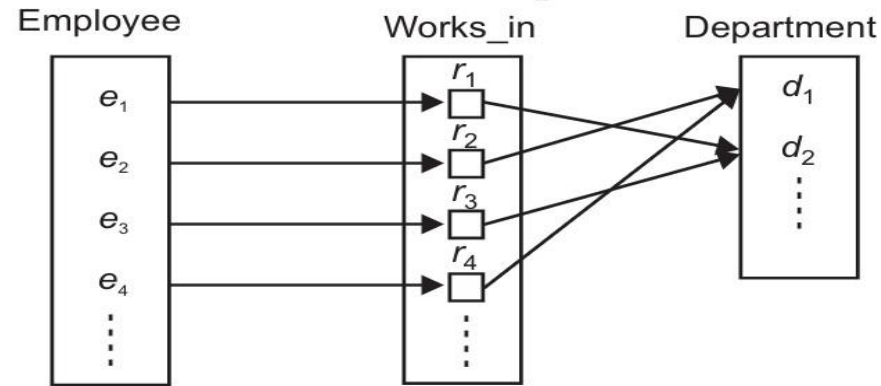
- **Descriptive Attributes :** Attributes of any relationship set are known as descriptive attributes. Works_in is a descriptive attribute.

Degree of Relationship Sets

Total number of entity sets participate in a relationship set is known as degree of that relationship set.

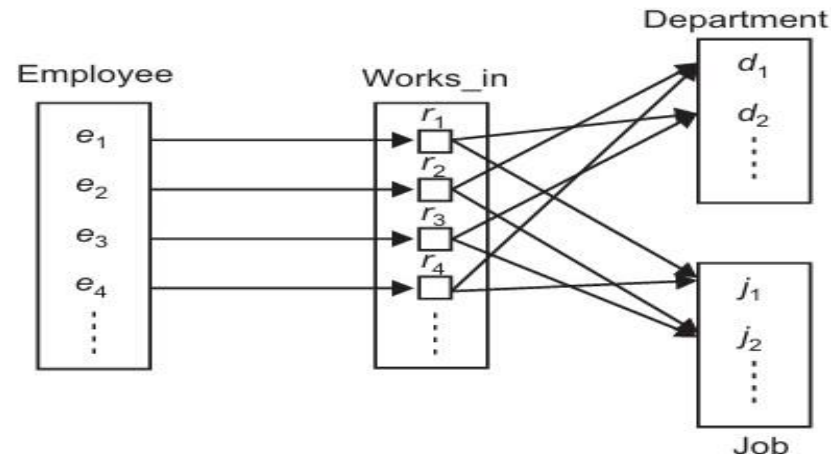
- i) **Binary Relationship Set:** A relationship set in which only two entity sets are involved is known as binary relationship set.

Ex.



- ii) **Ternary Relationship Set:** A relationship set in which three entity sets are involved is known as ternary relationship set or a relationship set having degree three.

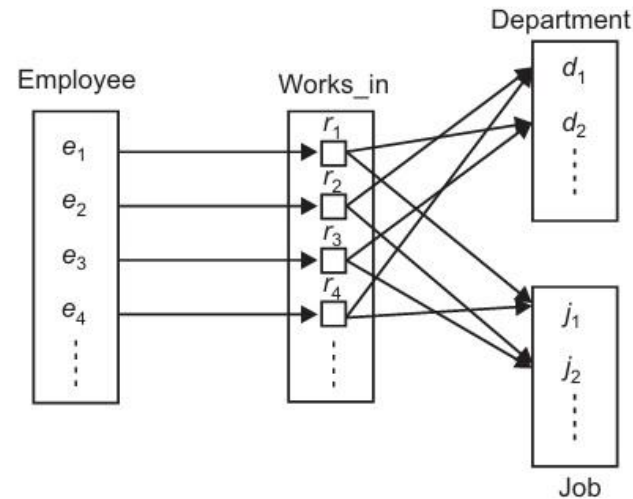
Ex.



Role and Recursive Relationship Set

Total number of entity sets participate in a relationship set is known as degree of that relationship set.

- **Role :** The function of any entity which it plays in relationship set is called that entity's role.
Ex. Employee plays the role of worker in his department in Figure 2.4.



- **Recursive Relationship Set :** When the same entity sets participate in same relationship set more than once with different roles each time, then this type of recursive relationship set is known as Recursive Relationship set.

For example, consider a relationship set `works_in` and two entity set `student` and `college`. A student who attends weekend classes in college as student may also be lecturer in that college. Then this person plays two roles (student, faculty) in same relationship set `work_in`.

Mapping Constrains

(5.a) Mapping Cardinalities (Cardinality Ratios):

It specifies the number of entities of an entity set that are associated with entities of another entity set through a relationship set.

□ Types of Mapping Cardinalities:

1. **One to One (1 : 1)** : An entity in X is associated with at most one entity in Y and an entity in Y is associated with at most one entity in X.

Ex. A country has only one president. Any person may be the president of at most one country

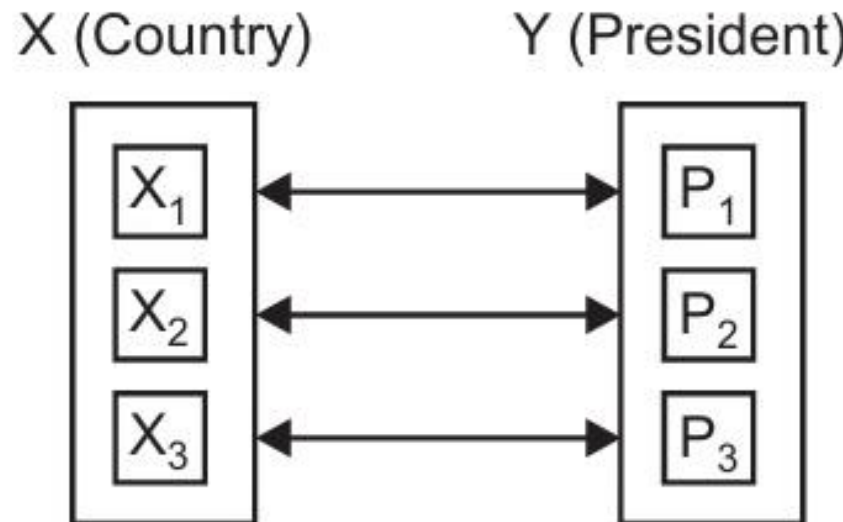


Figure: 1:1 relationship

2. One to Many (1 : N) : An entity in X is associated with any number of entities in Y. An entity in Y is associated with at most one entity in X.

Ex. A manager has many employees under it but an employee works under only one manager

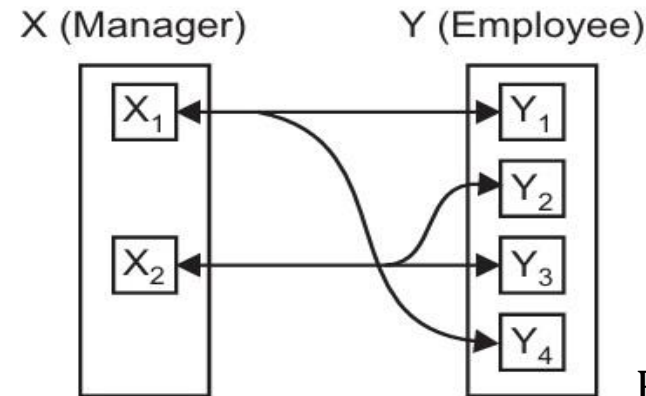


Figure: 1:N relationship

3. Many to One (N : 1) : An entity in X is associated with at most one entity in Y. An entity in Y is associated with any number of entities in X.

Ex. A employee can work on single project while any project can be assigned to more than one employee.

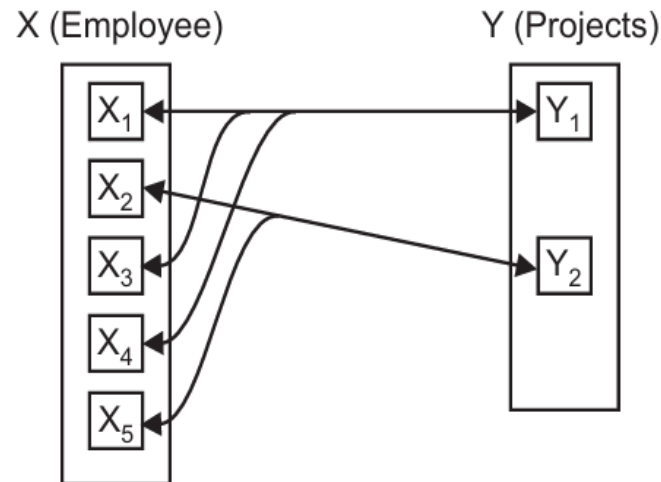


Figure: N :1 relationship

4. Many to Many (M : N) : An entity in X is associated with any number (zero or more) of entities in Y and vice versa.

Ex. A student can have more than one subject and one subject can be given to more than one student.

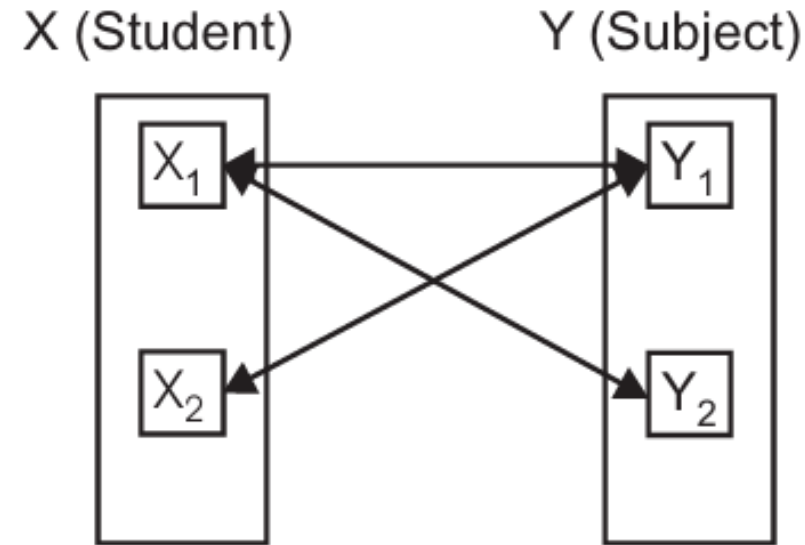


Figure: M: N relationship

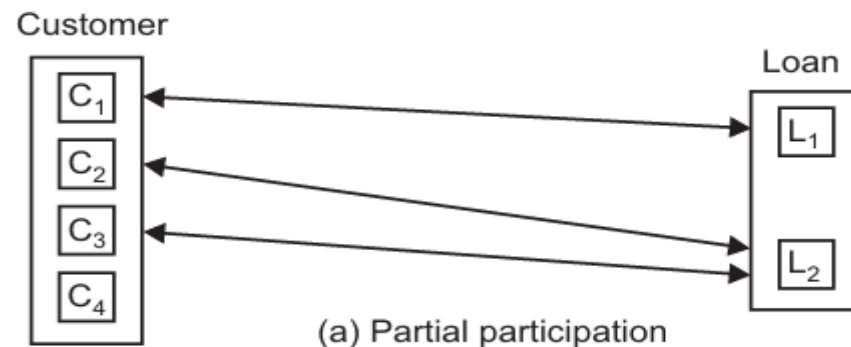
Participation Constraints :

It tells the participation of entity set in relationship sets.

□ There are two types of participations:

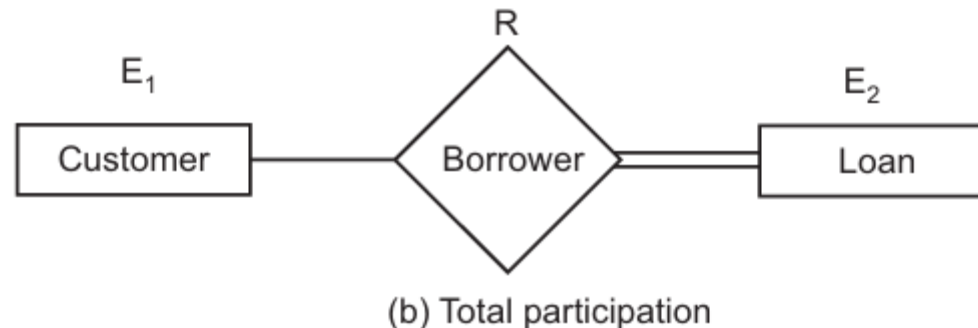
1. Partial participation : If only some entities from entity set E is participated in relationships in set R then it is known as Partial participation.

Ex. Every customer may or may not take the loan , so Customer entity set is partially participated.



2. Total participation : If every entity from entity set E is participated with at least one relation in relationship set R then it is known as Total participation.

Ex. Every loan is concerned with at least one customer of bank. So, loan entity set is totally participated.



Keys

- A key is an attribute or set of attributes that is used to uniquely identify data in entity sets.
- The attributes which are used as key are known as key attributes. Rest of all are known as Non-key attributes.

□ Types of Keys:

1. Super Key : A super key is a set of collection of one or more than one attributes that can identify data uniquely. Any entity set has more than one super key.

Ex.

Employee

Reg. No	ID	Name	Salary	Dept-ID
S1D	1	Mohan	1500	10
A25	2	Sohan	2000	30
33Z	3	Vikas	3000	20
Z4X	4	Madhu	1000	10
A5C	5	Sonal	5000	20

(a)

In entity set Employee, shown in Figure (a), Super Keys are

(a) (*ID, Name, Salary, Reg. No.*)

(b) (*ID, Name, Reg. No.*)

(c) (*ID*) etc.

All combinations can identify data uniquely.

Employee

Reg. No	ID	Name	Salary	Dept-ID
S1D	1	Mohan	1500	10
A25	2	Sohan	2000	30
33Z	3	Vikas	3000	20
Z4X	4	Madhu	1000	10
A5C	5	Sonal	5000	20

(a)

2. Candidate Key :

- The minimal super key is known as candidate key.
- Consider a super key and then take all of its proper subsets.
- If no one of the proper subsets are super key, then this super key is taken as candidate key.

Ex. *ID* and *Reg. No.* are candidate key.

3. Primary Key :

- An attribute which identifies data uniquely is known as Primary Key.
- The term Primary Key is used to denote Candidate key.

Ex. In entity set Employee, either *Reg. No.* is primary key or *ID* is primary key.

N.B. Any entity set can have **more than one Candidate key** but **only one Primary Key**.

4. Alternate Keys : All the candidate keys other than Primary Key are known as Alternate Keys.

Ex. If you take *ID* as Primary Key. Then, *Reg. No.* is an alternate key.

5. Secondary Key : An attribute or set of attributes which doesn't identify data uniquely but identifies a group of data is known as secondary key.

Ex. *Name, Salary and Department No.* are all secondary keys.

6. Foreign Key : A foreign key is an attribute in any entity set which is also a Primary Key in any other entity set.

Ex. *Dept-ID*: This is an attribute in entity set **Employee**(a) and also a primary key in entity set **Department**(b). Thus, it is a foreign key in Employee.

Employee

Reg. No	ID	Name	Salary	Dept-ID
S1D	1	Mohan	1500	10
A25	2	Sohan	2000	30
33Z	3	Vikas	3000	20
Z4X	4	Madhu	1000	10
A5C	5	Sonal	5000	20

(a)

Department

Dept-ID	Dept-Name
10	Sales
20	Marketing
30	Development

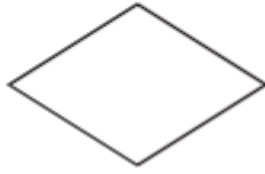
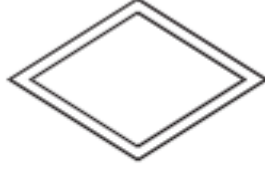
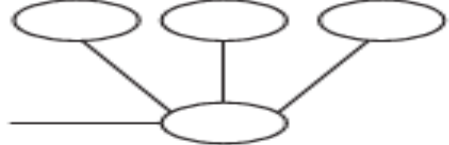
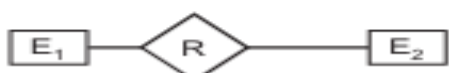
(b)

Entity Relationship Diagram(E-R Diagram) :

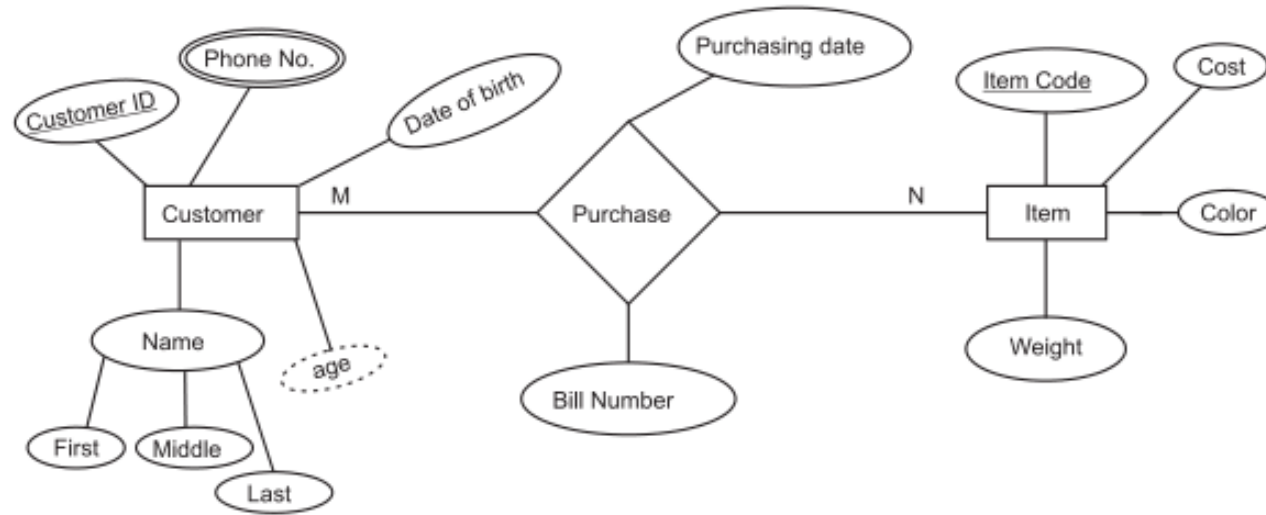
E-R diagrams represents the logical structure of a database. Symbols used in E-R diagrams are shown in Table 2.1.

Table 2.1. Symbols in E-R diagram

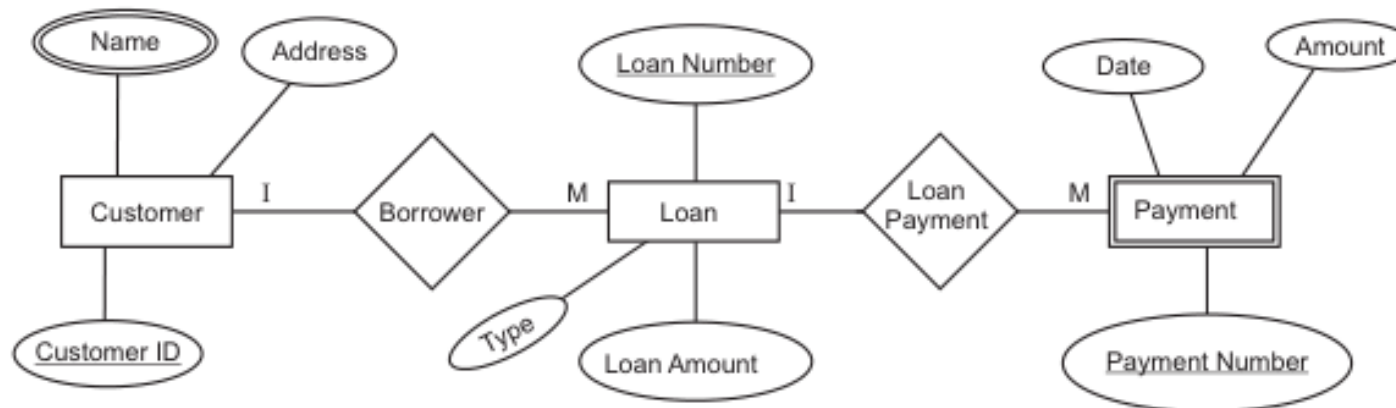
S.No.	Name of Symbol	Symbol	Meaning
1.	Rectangle		Entity Set (Strong)
2.	Double Rectangle		Entity Set (Weak)

3.	Ellipse		Attribute
4.	Diamond		Relationship Set
5.	Double Diamond		Identifying Relationship Type
6.	Double Ellipses		Multi-valued attributes
7.	Dashed Ellipses		Derived attributes
8.	Ellipse with line inside it		Key attribute
9.	Ellipse joined with other ellipses		Composite attributes
10.	Double lines		Total Participation
11.	Single line		Partial Participation
12.	Triangle		Specialization or Generalization

Ex-1. An E-R diagram having two entity sets, Customer and Item.



Ex-2. An E-R diagram with entities Customer, Loan and Payment.



❑ Advantages and Limitation of E-R model

Advantages:

- 1. Straightforward relation representation:** The relation representation of the database model using E-R diagram are relatively more straightforward than other models.
- 2. Mapping with relational model:** It can be easily mapped onto the relational model. The E-R diagrams used in the E-R model can easily be transformed into relational tables. The entities and attributes of E-R model can easily be transformed into relations (tables) and columns (fields) in a relational model.
- 3. Communication tool:** It is very simple and easy to understand with a minimum of training efforts required. Therefore, the model can be used by the database designer to communicate the design to the end user.
- 4. Design tool:** E-R model can also be used as a design plan by the database developer to implement a data model in specific database management software.
- 5. Easy conversion to other models:** E-R diagrams can be easily converted to a network or hierarchical data model.

6. Graphical representation: E-R model provides graphical and diagrammatical representation of various entities, their attributes and relationships between entities.

7. Easy to modify: Modifications to E-R diagram at later stage is relatively easier than in other models.

Limitation:

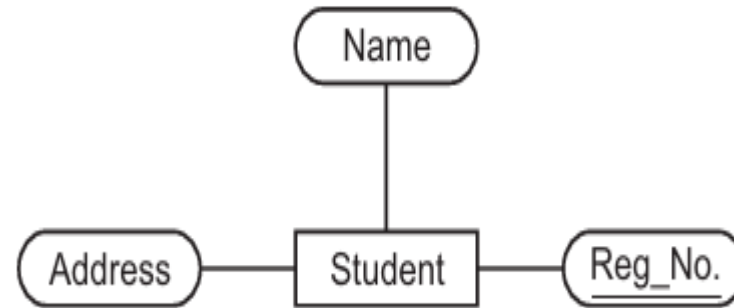
- ❑ E-R model cannot express relationships between relationships. In other words, E-R model is not capable to express relationship set between relationship sets.
- ❑ *This limitation can be overcome by using EER model.*

Types of Entity Sets

There are two types of entity sets as given below:

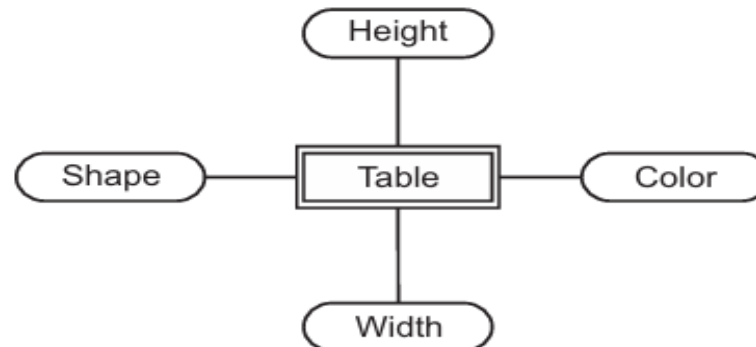
a) Strong Entity Sets: Entity set having any key attributes are known as Strong Entity sets.

Ex. The Figure shows strong entity set **Student** having key attribute *Reg_No.*



b) Weak Entity Sets: Entity sets having no key attributes are known as Weak Entity sets.

Ex. The Figure below shows weak entity set Table having no key attributes to identify the tuples uniquely.



N.B. A weak entity set is always associated with another entity set to make it meaningful.
*This another entity set is known as **Identifying entity set**.*

9. Enhanced Entity-Relationship (EER) Model

What is EER?

EER model is basically the enhanced version of E-R model which includes all the basic concepts of E-R model with capability to support **additional** semantic concepts of complex applications.

❑ Concepts of EER

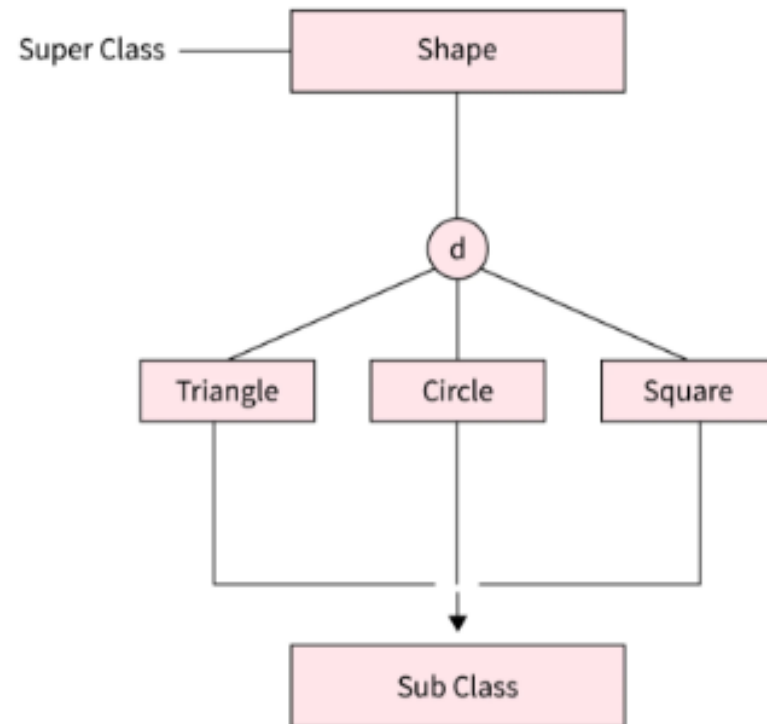
- Specialization
- Generalization
- Categorization

➤ Before discussing the concepts of specialization, generalization, and categorization two another entity types **superclass** (supertype) and **subclass** (subtype) are need to be described.

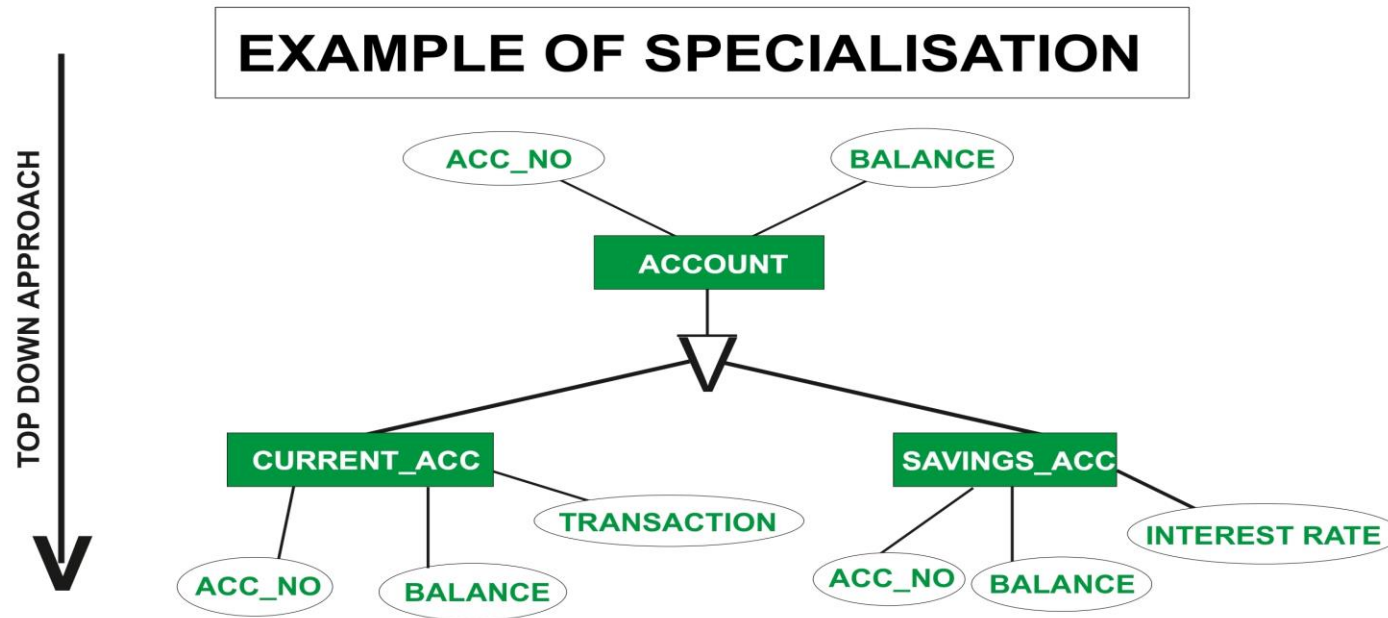
1. Superclass Entity Type (Supertype) : A superclass entity type is a generic entity type that includes one or more distinct subclasses that require to be represented in a data model.

2. Subclass Entity Type (Subtype) : A subclass entity type is a more specialized entity type that has a distinct role in the organization. (*A subclass is a member of superclass.*)

Ex.

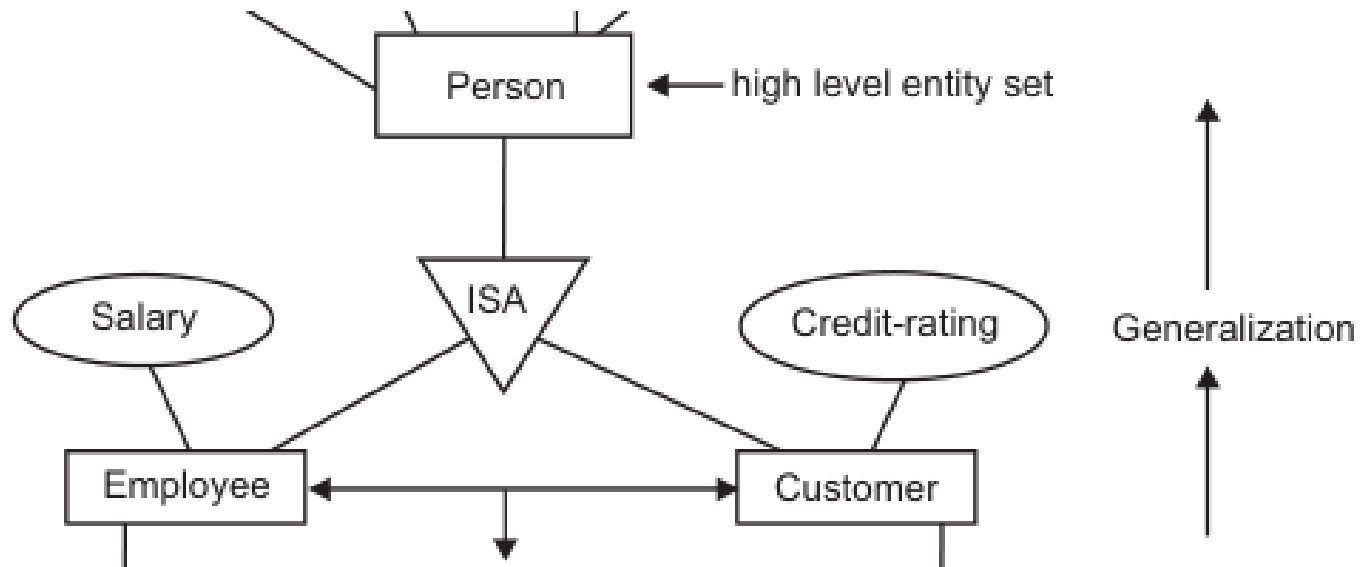


9.a) Specialization : Specialization is a process by which any existing entity set is divided into smaller entity sets according to the distinct or different nature of entities.



9.b)Generalization: Generalization is a process by which two or more entity sets can be combined into a single entity set by determining similarities between the entities.

Ex.



Here Employee and Customer are two entity sets. They have common feature that both are Person.

❑ Specialization and Generalization Constraints

The following constraints are applied on specialization and generalization to capture important business rules of the relationships in an enterprise.

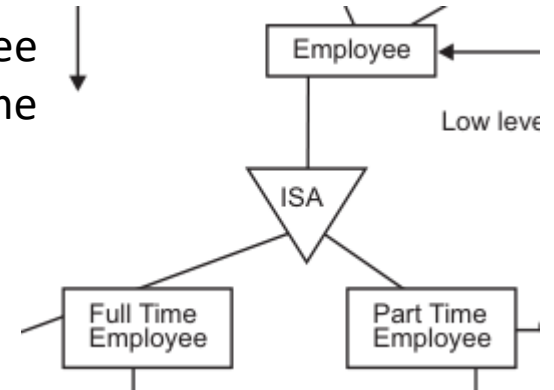
❖ There are Two types of constraints :

1. Participation Constraints: *Discussed in (5.b)*

2. Disjoint Constraints : According to Disjoint constraint if the subclasses of a specialization/generalization are disjoint then an entity can be a member of only one subclass of that specialization/generalization.

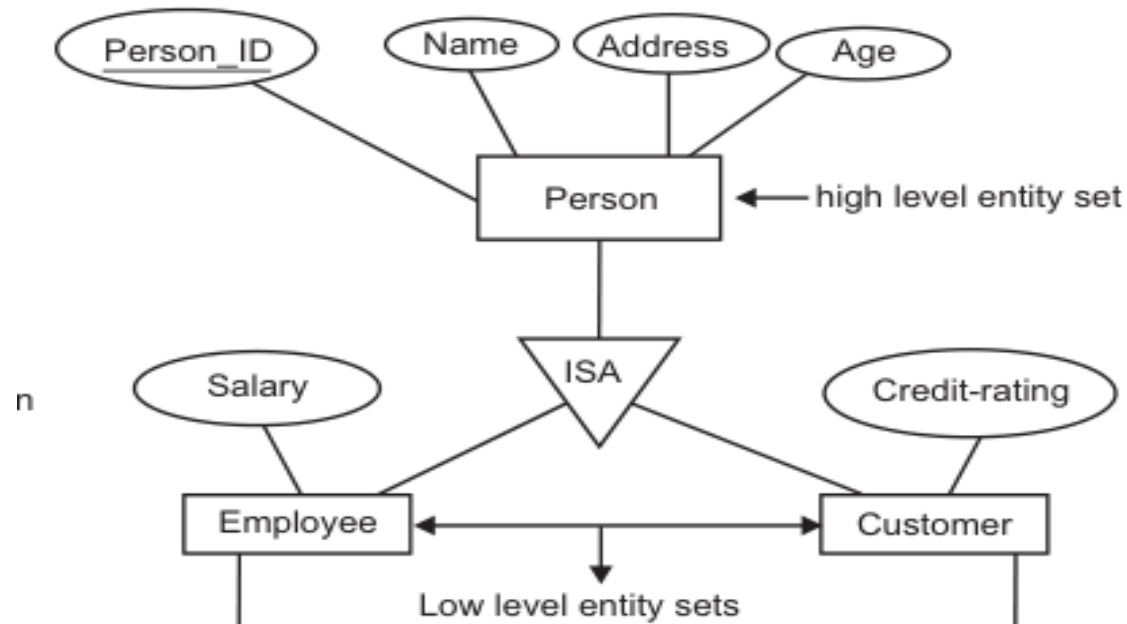
Ex.

Suppose any employee 'Martin' works as part time employee for Bank then it can only belong to subclass 'Part Time Employee'.



9.c) Attribute Inheritance: Attribute Inheritance is a process by which lower-level entity set inherits (or taken) some properties of its higher-level entity set.

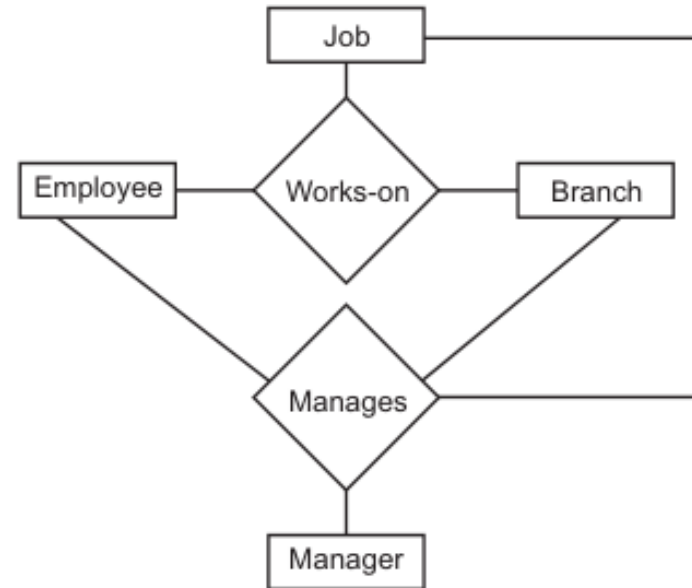
Ex.



Here entity sets **Employee** and **Customer** inherits attributes *Person_ID*, *Name*, *Address*, *Age* from **Person** entity set.

9.d) Aggregation: Aggregation in DBMS is process of combining two or more entities to form a more meaningful entity.

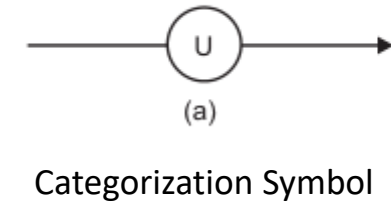
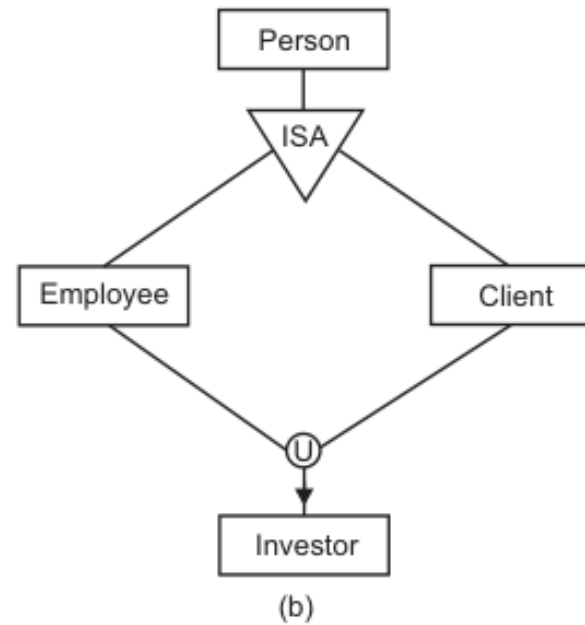
Ex.



In this example, the **Employee** and **Branch** alone do not make sense and are not useful. However, when they are combined through the aggregation process, a more meaningful entity **Manager** is formed. This New entity can perform better and provide more meaningful information.

9.e) Categorization: The subclass having more than one superclass is known as category and the process of defining a category is known as Categorization.

Ex.



In a bank, a **person** can be either an **employee** or a **client** and both of them may be **investor**. So, here Subclass **employee** and **client** act as disjoint **Superclasses** and subclass **investor** acts as category.

10. Reduction of an E-R and EER Diagram into tables

To represent the database in tabular form, E-R diagrams have to be reduced in tables. For each entity set, make different table and for each relationship set make a different table.

10.1 Reduction of Strong Entity Sets into Tables

For a strong entity set E with attributes a1, a2, ..., an,

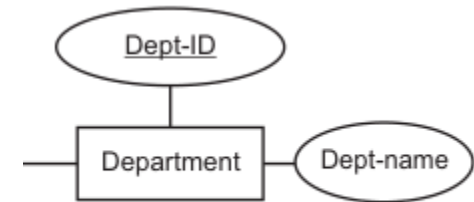
- Step-01: Make a table having same name as of entity set E
- Step-02: Make n number of columns where 'number of columns' = 'number of attributes' and columns name = entity set name.

Ex.

Consider a strong entity set **Department** with two attributes *Dept-ID* and *Dept-name*. Then the corresponding table will be

Department

Dept-ID	Dept-name
10	Sales
20	Development
30	Testing
40	Accounts

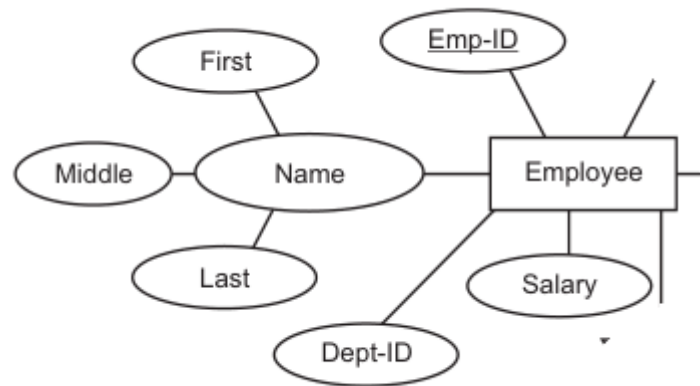


10.2 Reduction of Composite Attributes

For a composite attribute, create a separate column for each component attribute or parts of composite attributes.

Ex.

Consider the example shown in Figure 2.20. The *Name* is a composite attribute with three component attributes *First*, *Middle* and *Last*. So, make three columns **First-name**, **Middle-name** and **Last-name**. Then the corresponding table shown in Figure 2.22



EID-ID	First-name	Middle-name	Last-name	Salary	Dept-ID
A12	Deepak	Kumar	Goyal	15,000	10
S50	Shivi	—	Goyal	75,000	20
51C	Anu	—	Parmar	8,000	10
67B	Ravi	—	—	5,000	40

FIGURE 2.22. The employee table (Reduction of composite attributes).

Figure 2.20. E-R Diagram of entity set Employee

10.3 Reduction of Multi-valued Attributes

For multi-valued attributes, make a separate table with columns C1 which represent the **primary key** of entity set or relationship set and with columns C2 which represent the **multi-valued** attributes. Rows are equal to total number of values of that attribute.

Ex.

Consider Figure 2.20 in which *Phone-No.* is multi-valued attribute. So, make a table with two columns, one is **Emp-ID** (primary key of Employee) and second is **Phone-No.** (multi-valued attribute). Give any name to that table. The table is shown in Figure 2.22.

If any employee has two phone numbers, then it is possible to make two different entries in table and so on.

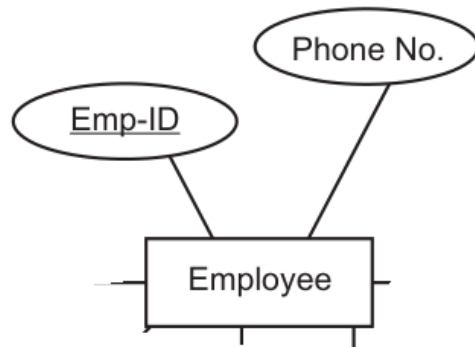


Figure 2.20. E-R Diagram of entity set Employee

Emp-ID	Phone-No.
A-12	23896
A-12	23897
51-C	38976
51-C	23551
51-C	98941
67-B	23999

FIGURE 2.23. The phone-number table (Reduction of multi-valued attributes).

10.4 Reduction of Weak Entity Sets

Let A be the weak entity set and B be the strong entity set on which A depends. Then, it is possible to make a table with table name same as of Weak Entity Set having **columns equal to the attributes of Weak Entity set plus Primary Key of the Strong Entity set** on which Weak Entity Set depends.

Ex.

Consider the E-R diagram shown in Figure 2.24, in which **Payment** is a **Weak entity set** that depends upon **Strong entity set Loan**(*has primary key*). So, make a table with table name Payment having four columns as shown in Figure 2.25.

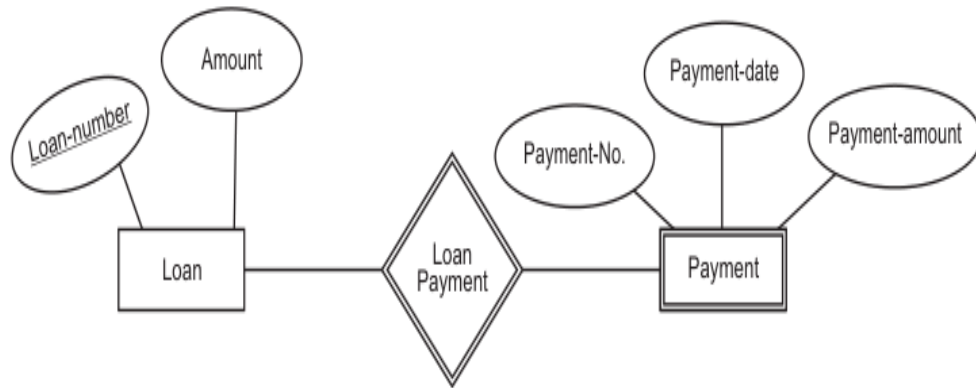


FIGURE 2.24. E-R diagram of weak entity set payment.

Payment

Loan-number	Payment-No.	Payment-date	Payment-amount
E-12	2	19-2-2004	6912
C-55	5	31-1-2005	5000
H-96	11	2-2-2005	2000
P-77	2	6-9-2005	2500

FIGURE 2.25. The payment table (Reduction of weak entity set).

10.5 Reduction of Relationship Sets

Let **R** be the relationship set and **E1, E2, ..., EN** be the entity sets participating in **R**. Make a table with table name same as of **Relationship Set** having columns equal to number of **attributes in relationship set** (descriptive attributes) and **primary keys** of all participating entity sets.

Ex.

Consider the ER diagram shown in Figure 2.20, having relationship set **Works-in** having two participating entity sets, **Employee** and **Department**. The corresponding table is shown in Figure 2.26.

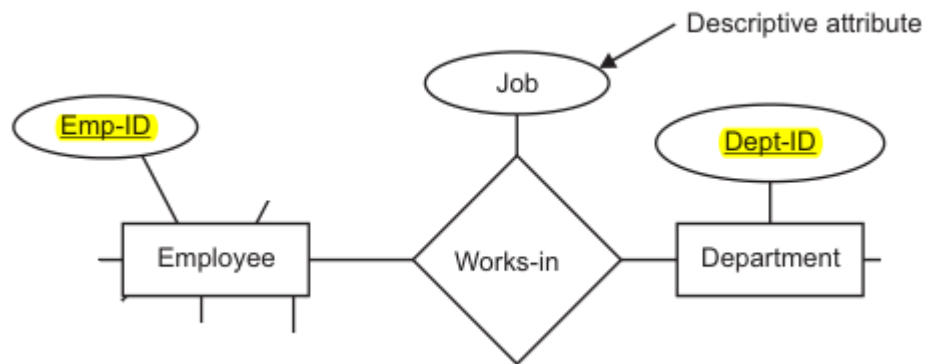


Figure 2.20. E-R Diagram

Works-in

Emp-ID	Dept-ID	Job
S-50	20	Engineer
A-12	10	Salesman
51-C	10	Salesman
67-B	40	Accountant

FIGURE 2.26. The works-in table (Reduction of relationship sets).

10.5.i) Redundant Tables

Redundant tables in DBMS refer to the situation where the same information is stored in multiple tables, leading to unnecessary repetition and increased storage requirements. This can occur when reducing aggregation into tables, particularly when dealing with relationship sets between weak and strong entity sets.

10.5.ii) Combination of Tables

Combination of tables refers to the process of merging or joining two or more tables based on common column or attribute. This is typically done to combine data from multiple tables into single table for easier querying and analysis.

10.6 Reduction of Generalization

In generalizations, higher level entity sets and lower level entity sets are considered. Make a table for higher level entity set with all its attributes. For lower level entity set, make a table with all its attributes with primary key attributes of its higher level entity set.

Ex.

Consider E-R diagram shown in Figure 2.20, in which **Employee** is high level entity set and **Full Time Employee** and **Part Time Employee** are two lower level entity sets. So, make three tables as given below:

- Employee [**Emp-ID**, Dept-ID, First-Name, Middle-Name, Last-Name, Salary]
- Full Time Employee [**Emp-ID**, Qualification]
- Part Time Employee [**Emp-ID**, Hours-Worked]

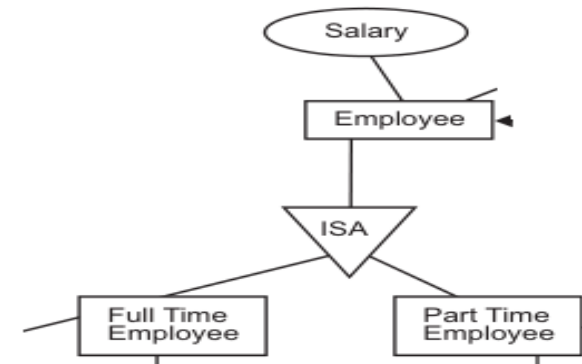


Figure 2.20. E-R Diagram of entity set Employee

10.7 Reduction of Aggregation

Reduction of aggregation into tables is simple. Consider the E-R diagram shown in Figure 2.19. For all entity sets, make tables as discussed earlier. For making tables for relationship sets, consider the same approach as discussed earlier.

Ex.

Take an example of relationship set *Manages*. Make a table *manages* with all descriptive attributes, primary key of entity set *Manager* and the relationship set *works-on*.

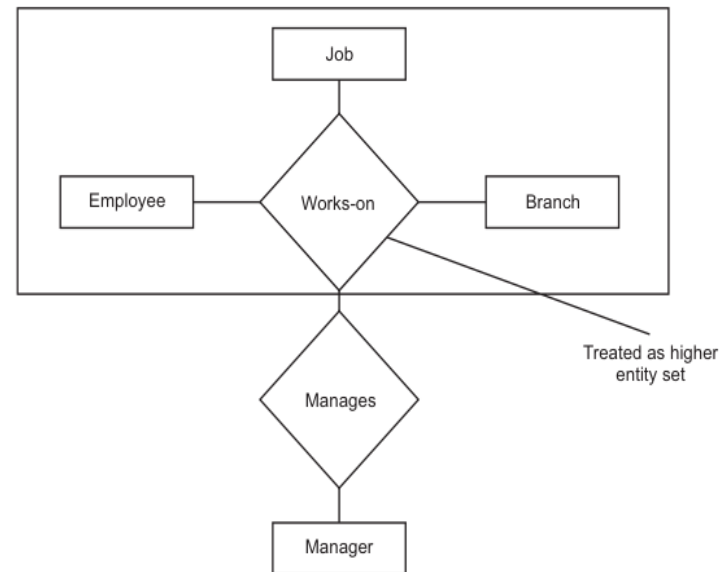


FIGURE 2.19. E-R model with aggregation.

❏ Problem Exercise

❖ Problem 1:

Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient, a log of various tests and examinations conducted. Construct the appropriate tables for this E-R diagram and list the tables with their attributes, primary key and foreign keys.

Solution: The E-R diagram is shown in Figure 2.28.

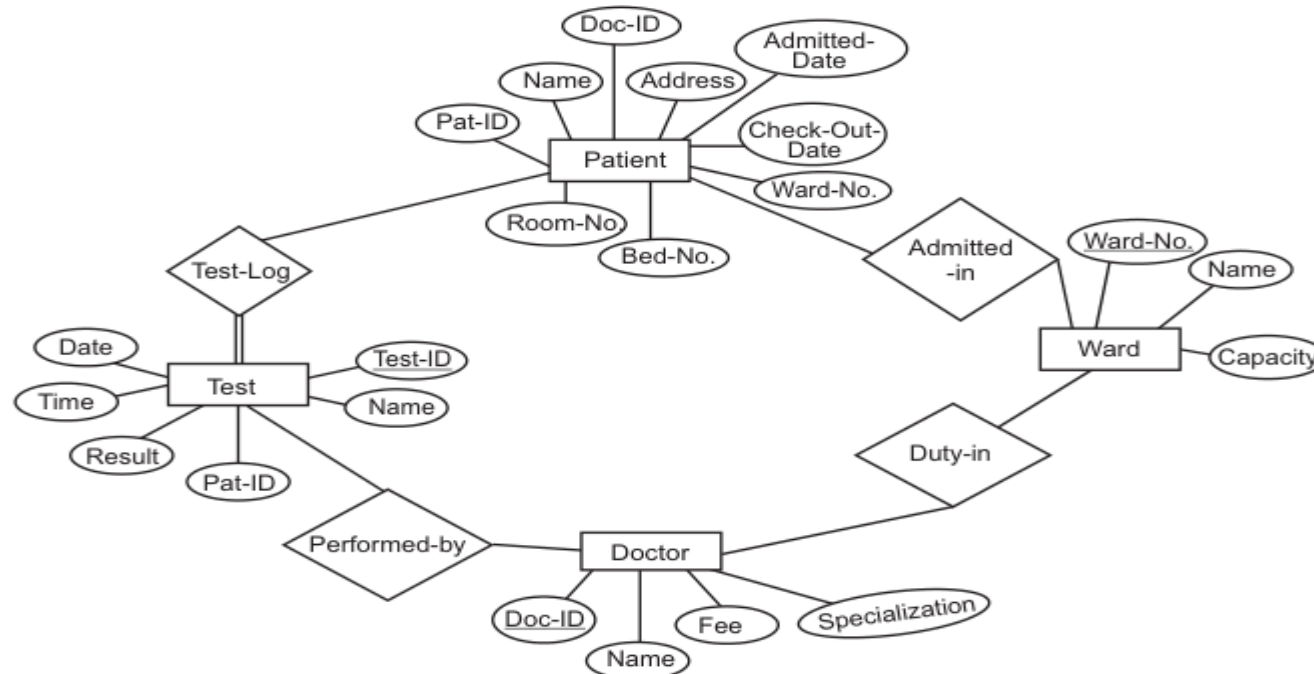


FIGURE 2.28. E-R diagram of hospital.

The Tables are as follows :

1. Patient (*Pat-ID, name, address, admitted-date, check-out-date, room-no., bed-no., ward-no, doc-ID*)
2. Ward (*Ward-no., name, capacity*)
3. Doctor (*Doc-ID, name, fee, specialization*)
4. Test (*Test-ID, name, date, time, result, Pat-ID*)
5. Primary key is shown by _____.
6. Foreign key is shown by _____.

Thank You